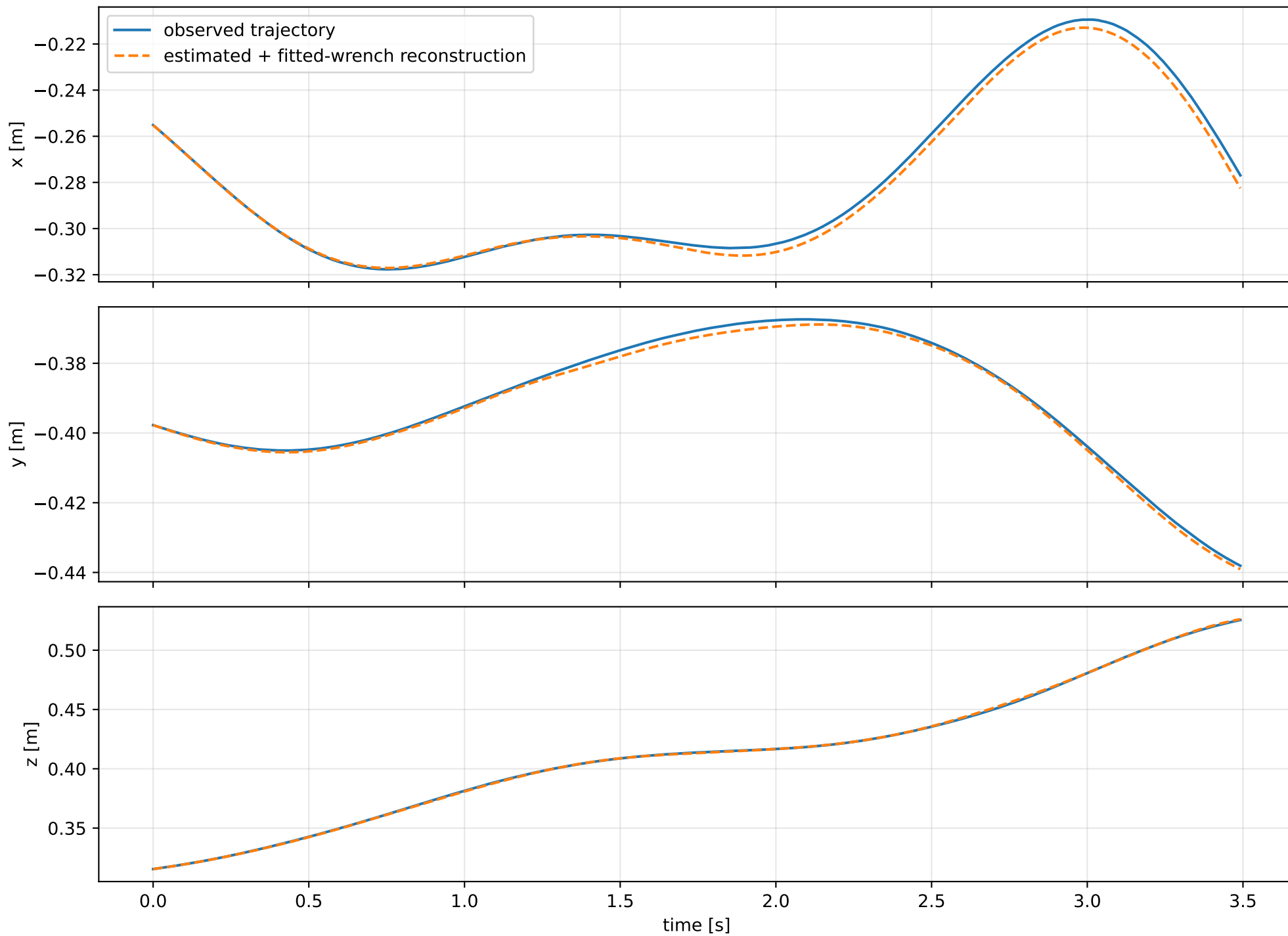
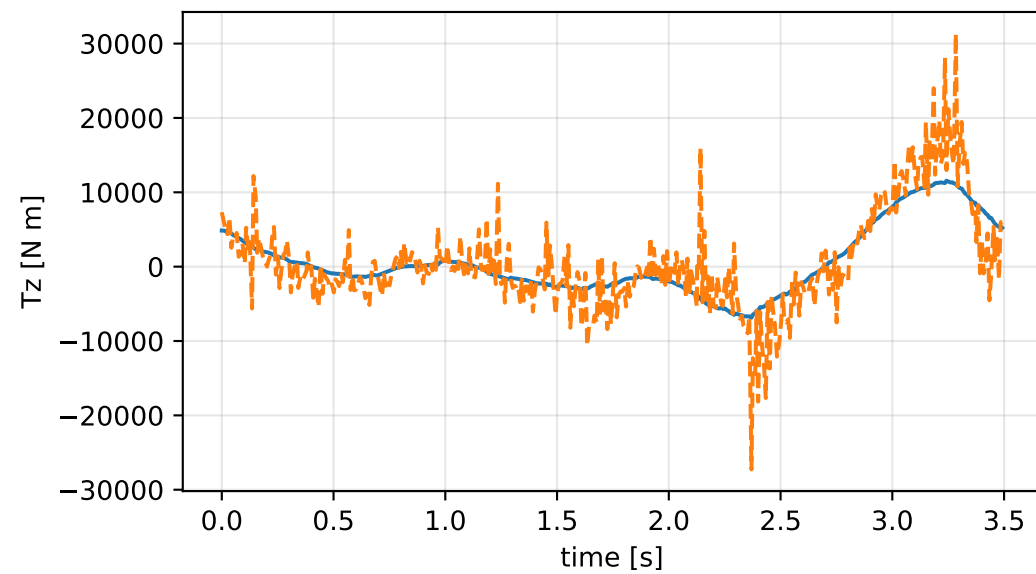
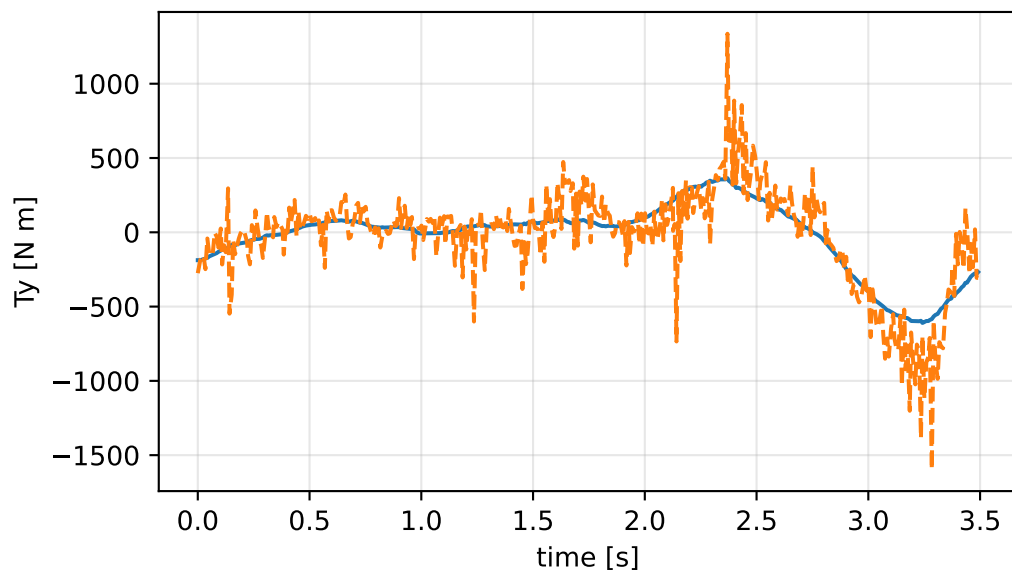
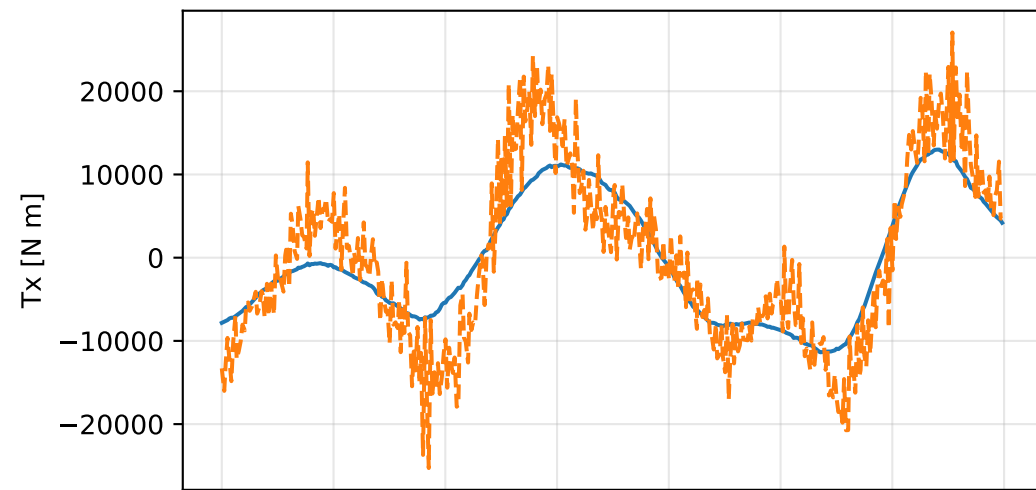
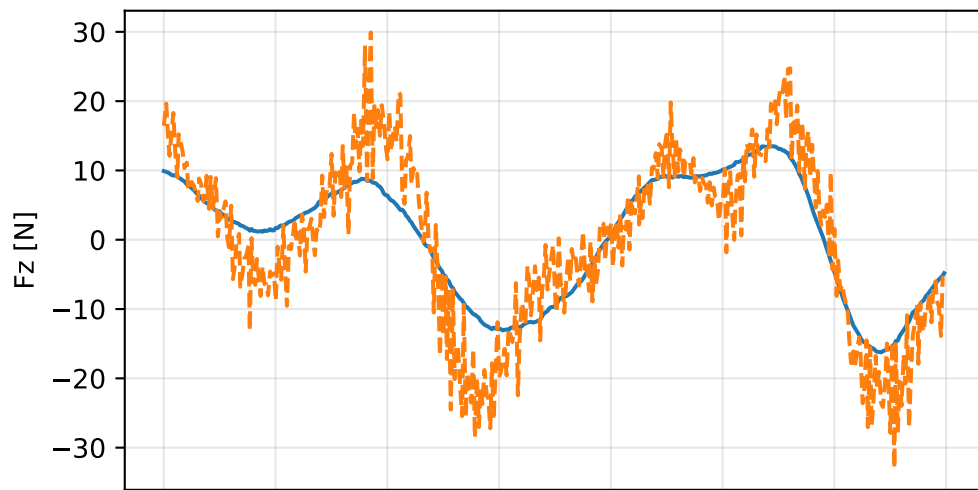
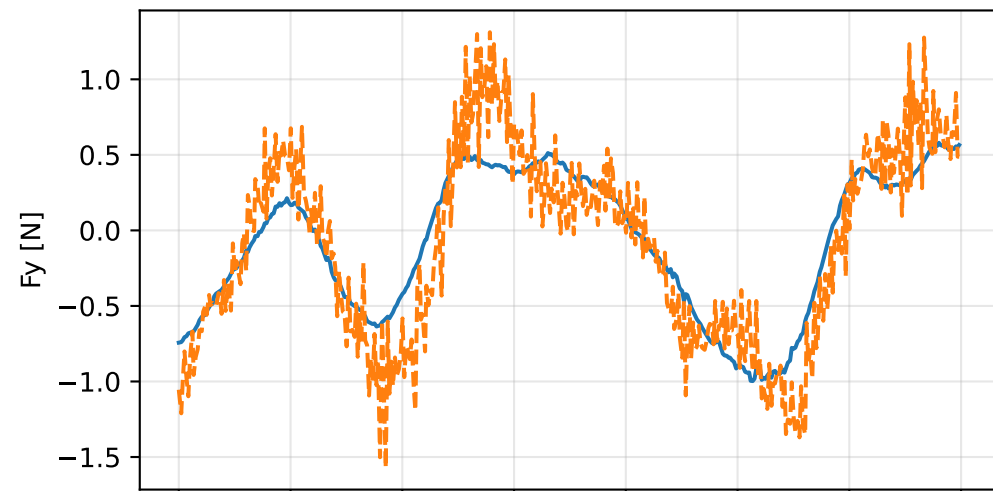
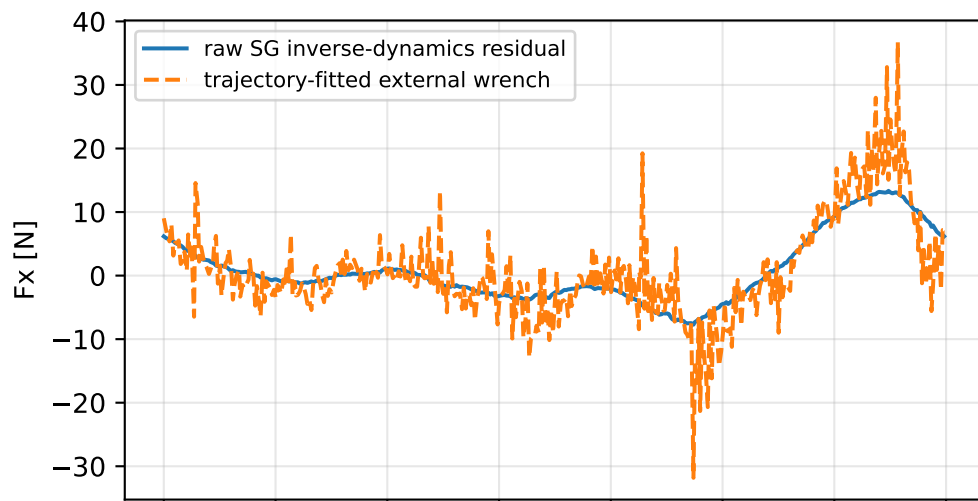


sweep\_sg\_window\_covariance\_005: trajectory

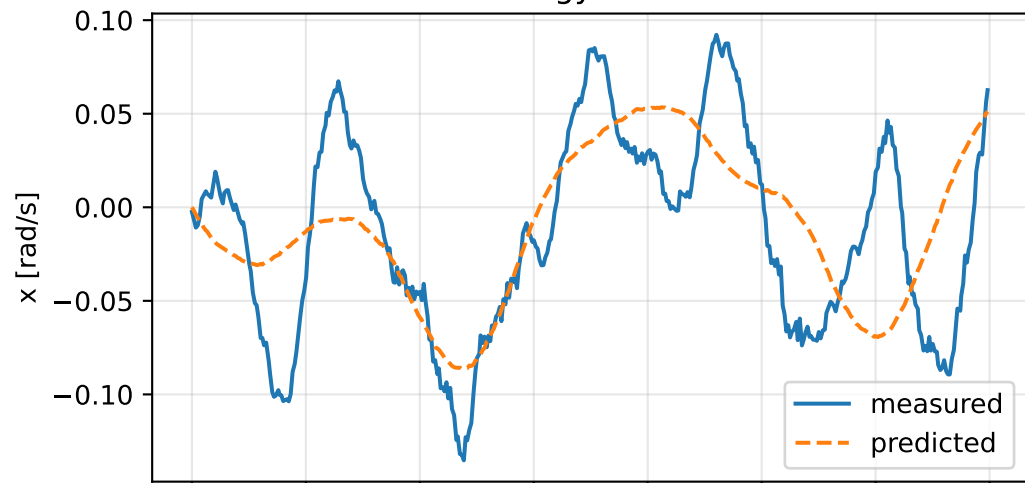


sweep\_sg\_window\_covariance\_005: residual wrench

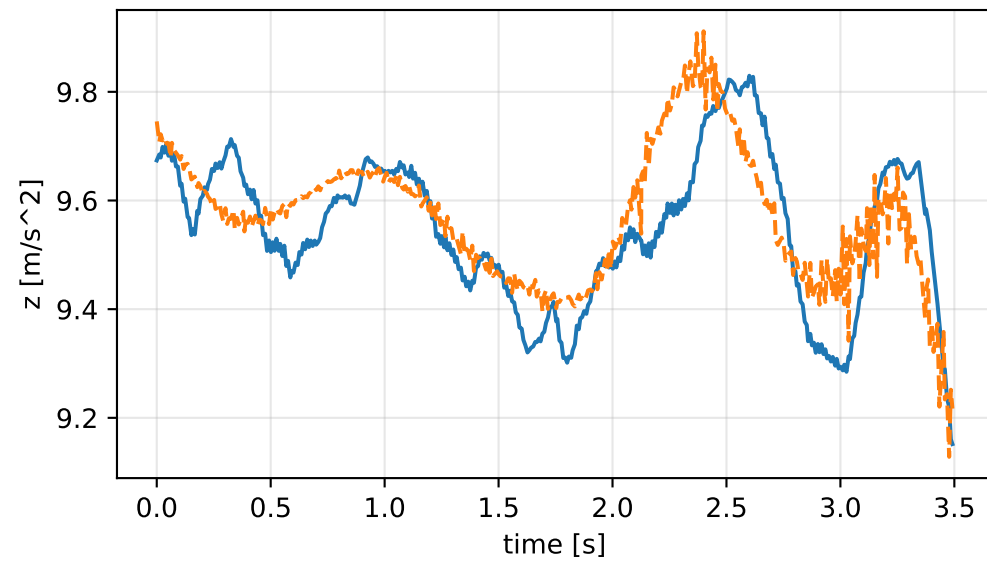
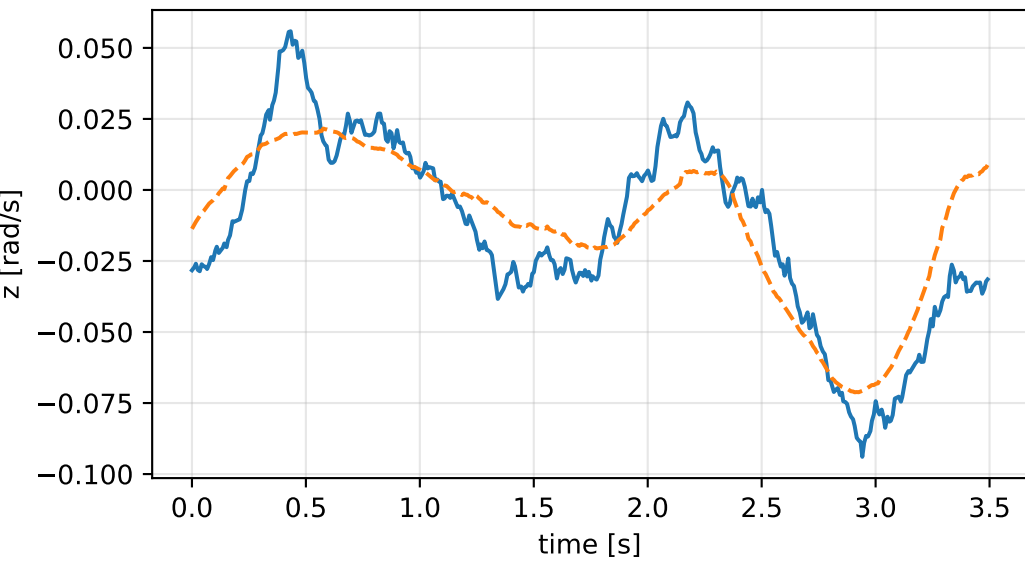
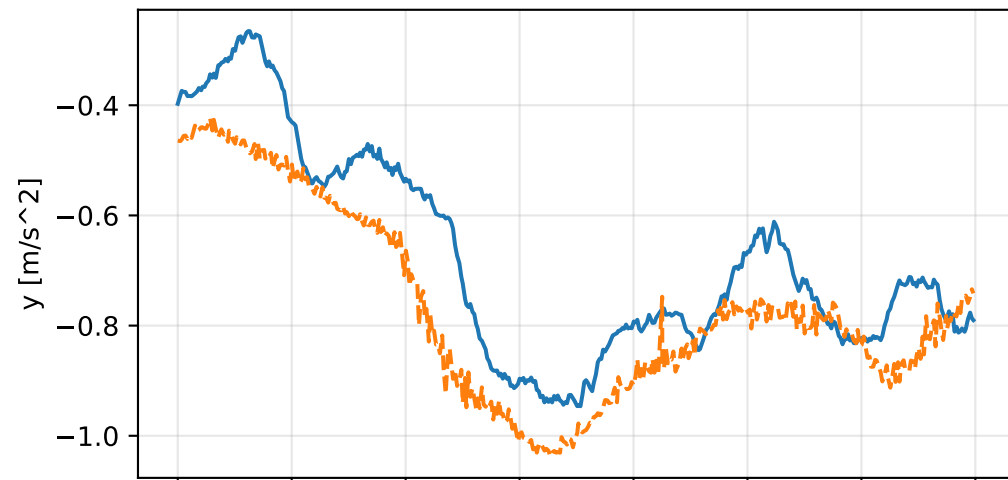
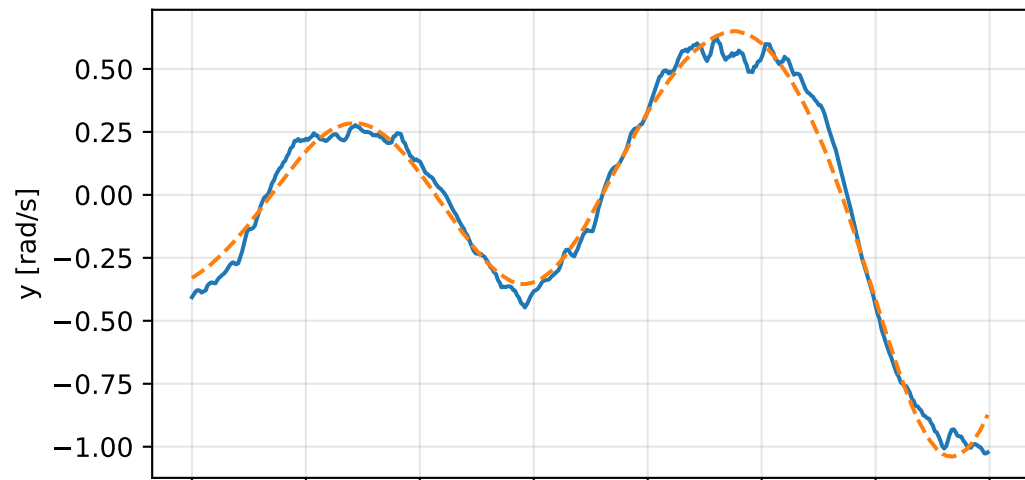
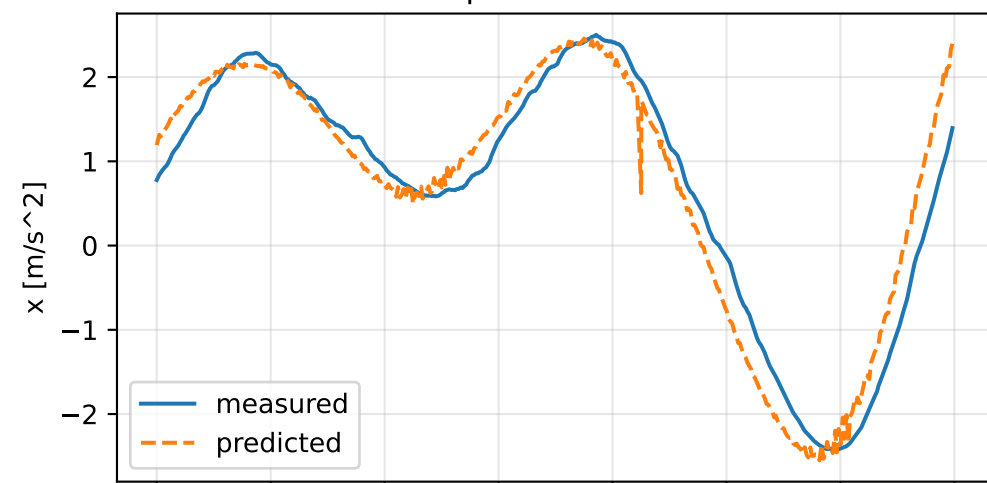


## sweep\_sg\_window\_covariance\_005: measured sensor comparison

gyro



specific force



## sweep\_sg\_window\_covariance\_\_005: nominal -> estimated parameters

Case: sweep\_sg\_window\_covariance\_\_005  
status: completed (all stages completed)

mass [kg]            nominal 2.35159759    estimated 2.37664381    delta 0.0250462146

inertia [kg m^2] (nominal -> estimated; delta)

[ 6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [ 5.7423935e+04 -2.9994466e+02 -1.4168751e+01]    d=[ 5.7423870e+04 -2.9994466e+02 -1.4168751e+01]  
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [ -299.9446613    128.8152086 -2699.2910247]    d=[ -299.9446605    128.7502086 -2699.2910247]  
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-1.4168751e+01 -2.6992910e+03 5.7298318e+04]    d=[ -1.4168770e+01 -2.6992910e+03 5.7298318e+04]

CoG [m]            [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [ -0.2004843 -28.5426161 -1.3198936]    d=[ -0.1984596 -28.5426161 -1.3198936]

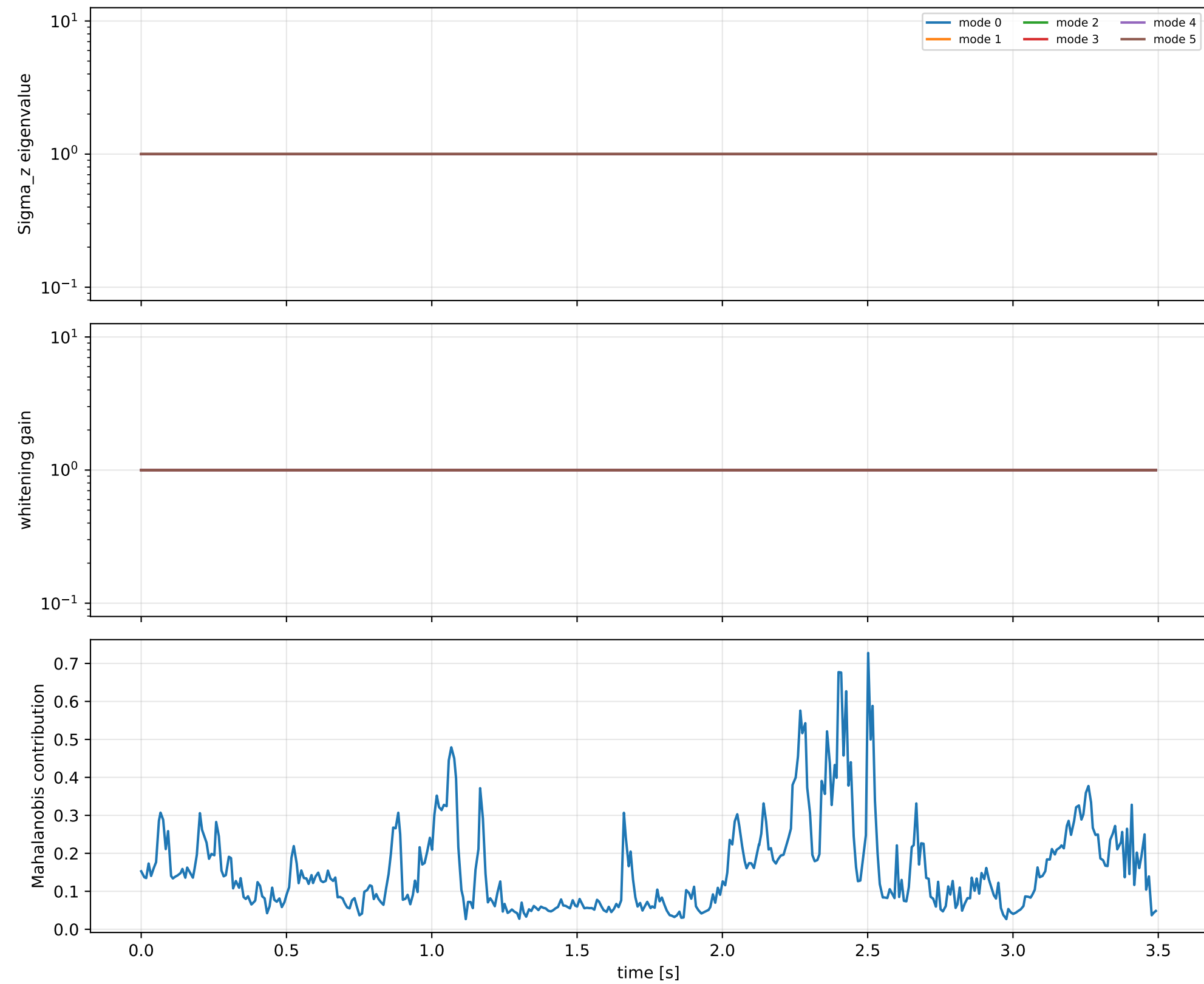
rotor force effectiveness

[1. 1. 1. 1.] -> [1.7012300e+00 1.3056765e+00 2.0477444e-05 3.0607117e-01]    d=[ 0.70123    0.3056765 -0.9999795 -0.6939288]

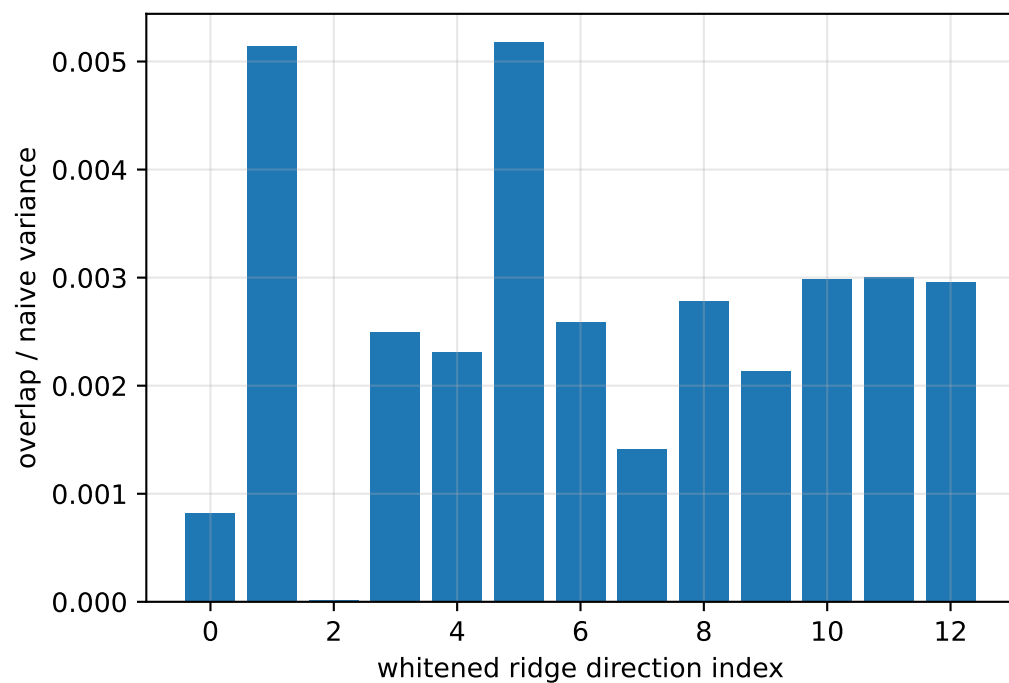
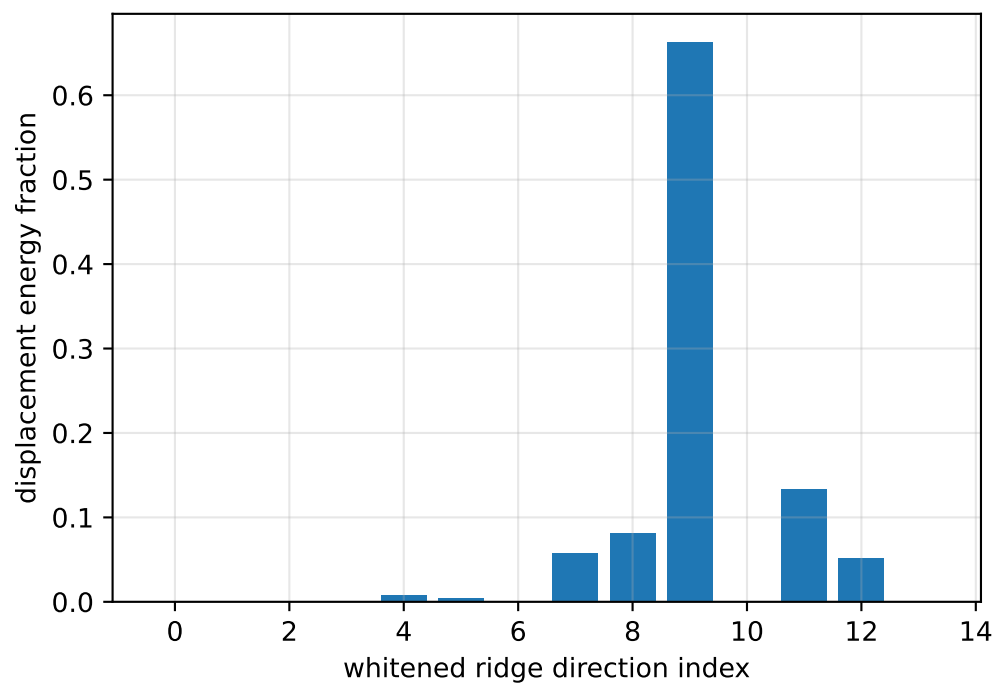
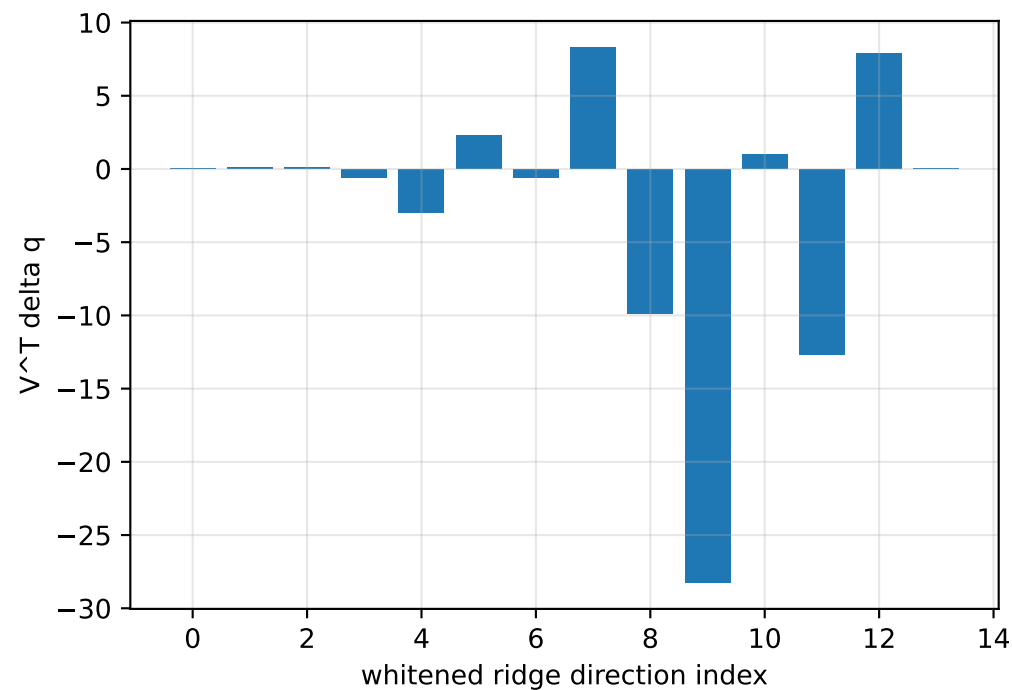
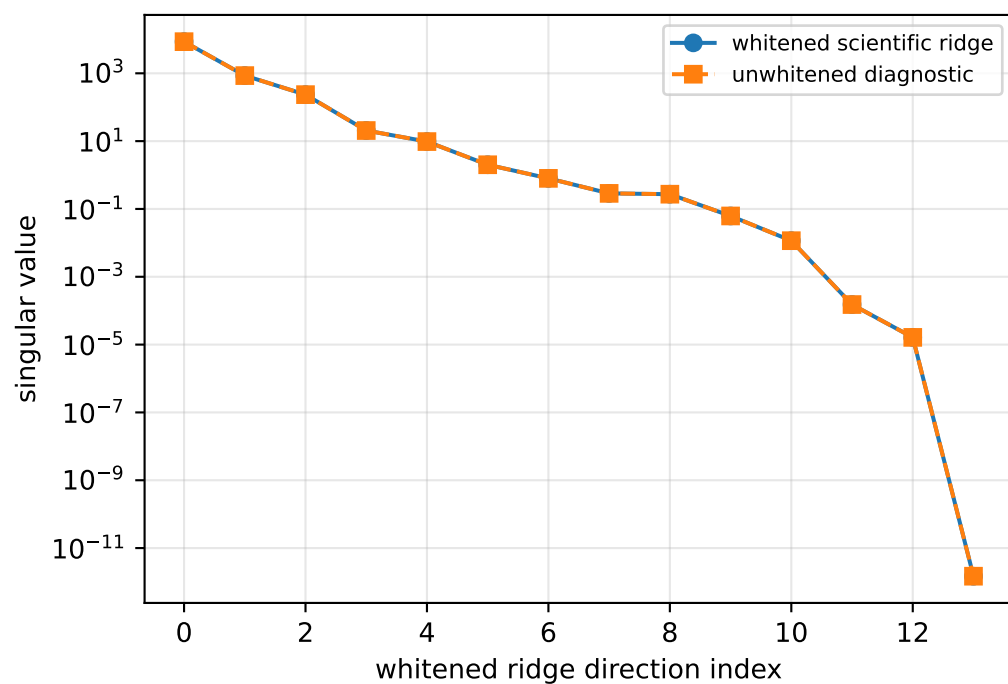
rotor lag [s] 0.2

gimbal lag [s] 0.00991439819

sweep\_sg\_window\_covariance\_\_005: covariance weighting

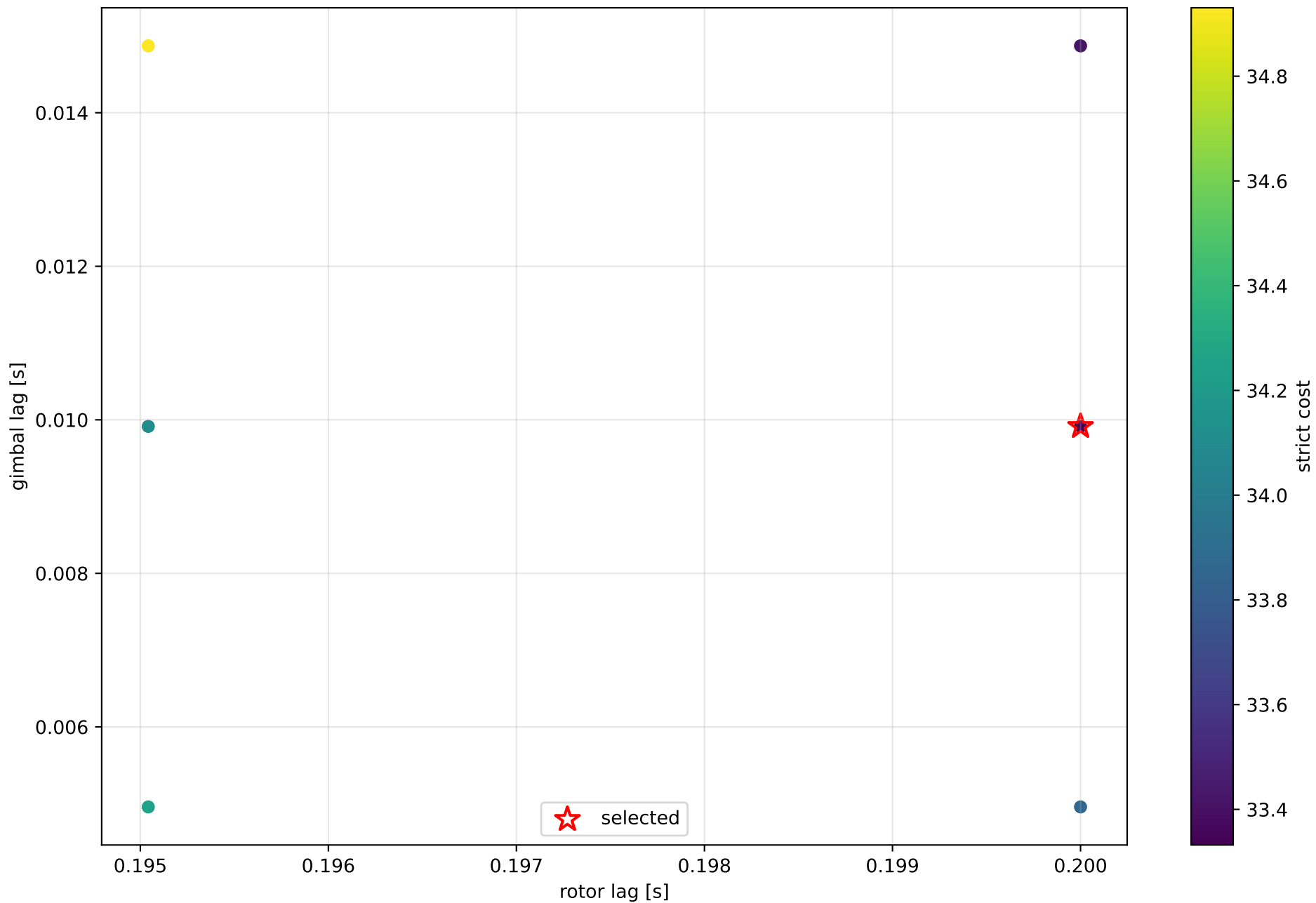


sweep\_sg\_window\_covariance\_005: parameter ridge



exact scale gauge:  $||j \ v\_scale||=3.06642e-09$ , rank=13, nullity=1

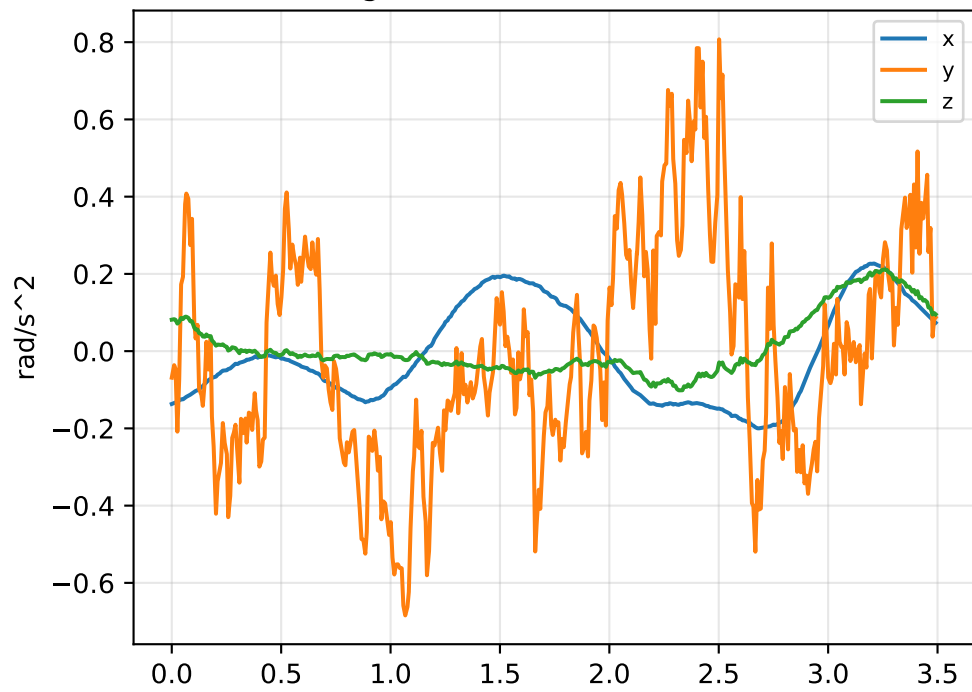
sweep\_sg\_window\_covariance\_\_005: strict lag candidates



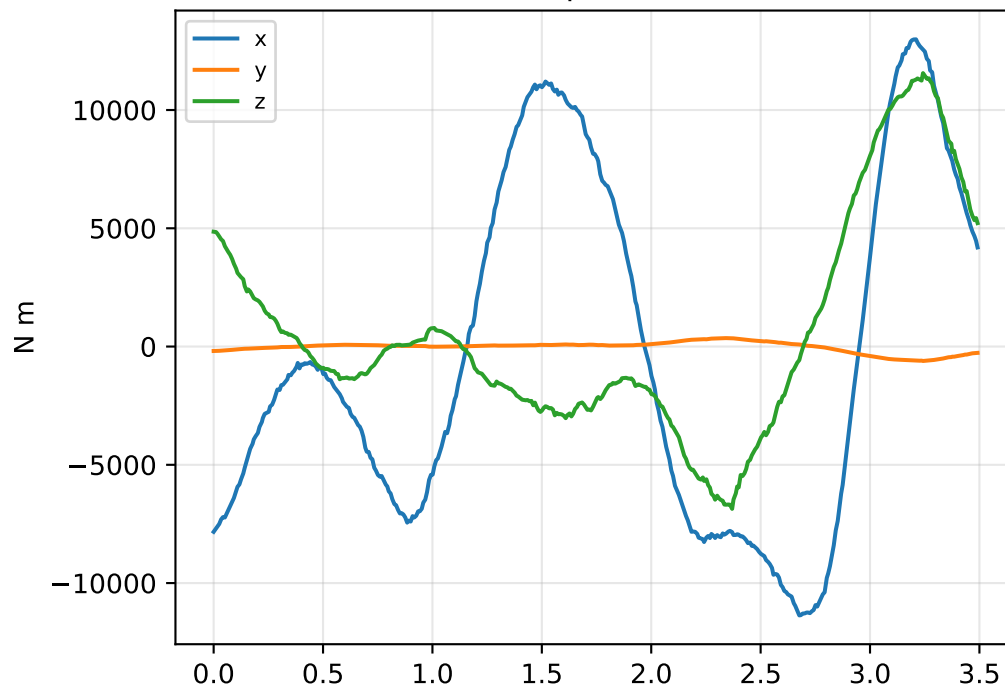
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

# sweep\_sg\_window\_covariance\_005: acceleration/wrench closure

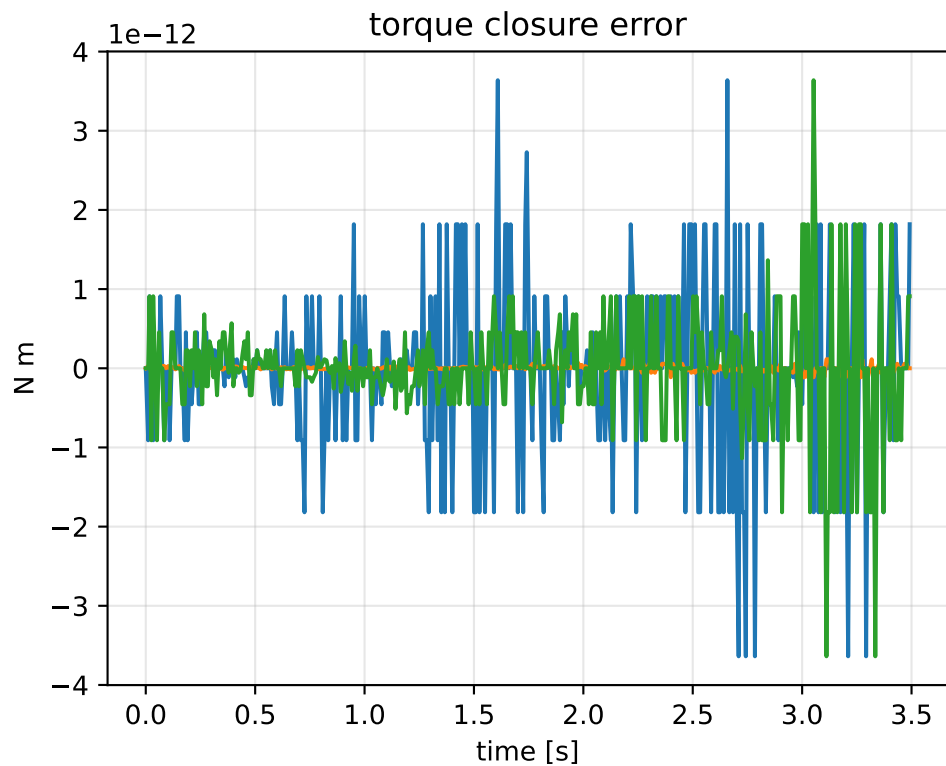
## angular-acceleration residual



## raw torque residual



## torque closure error



```
principal moments [kg m^2]
nominal: [0.064953 0.065 0.128992]
estimated: [7.927830e-02 5.742545e+04 5.742553e+04]
estimated / nominal: [1.220555e+00 8.834676e+05 4.451864e+05]

force closure max/rms: 7.10543e-15 / 1.66525e-15
torque closure max/rms: 3.63798e-12 / 6.9792e-13
```